

XIYUN LIU (LILY)

GRADUATE SOFTWARE ENGINEER | FULL-STACK DEVELOPMENT

Auckland, NZ | 027 489 3879 | xliu713@aucklanduni.ac.nz | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Graduate Software Engineer with three years of professional frontend development experience, currently completing a Master of Information Technology at the University of Auckland. Passionate about building scalable web applications and solving practical problems through systematic thinking. Founder of a large-scale animal rescue organisation, demonstrating leadership, resilience, operational decision-making and long-term ownership. Proven ability to take ownership, lead cross-functional initiatives and deliver user-focused solutions from concept to production.

KEY COMPETENCIES

Methodical Problem Solving • Ownership • Leadership • Collaboration • Adaptability

Known for combining structured problem solving with strong ownership and collaborative leadership to deliver practical software solutions.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, SQL, HTML5, CSS3
 - **Frontend:** React, React Native, Vue.js, Svelte, Zustand, TanStack Query, Tailwind CSS
 - **Backend:** Node.js, NestJS, Express.js, REST APIs, Socket.IO
 - **Data & Services:** PostgreSQL, MySQL, MongoDB, Prisma, Stripe
 - **Tools & Practices:** Git, GitHub, Docker, Agile/Scrum, Code Reviews, Unit/E2E Testing
-

PROJECTS

ChromaWalk — Colour Exploration App

<https://github.com/xiyunliu610/ChromaWalk>

A location-based social web app for discovering, capturing and sharing colours through collaborative challenges.

Tech Stack: React, TypeScript, Node.js, Express, MongoDB, Socket.IO

Key Contributions

- Delivered a production-style location-based social web application within a six-person Agile team, owning frontend development and collaborating closely with backend engineers to ship iterative releases.
- Designed and implemented responsive React interfaces with a strong focus on usability, maintainability and reusable component architecture, using Zustand and TanStack Query for state and data fetching.
- Collaborated closely with backend developers to build real-time chat and live progress updates, designing the communication layer around shared TypeScript interfaces.
- Integrated Google OAuth, Google Maps location selection and client-side image compression, and upheld code quality through version control and code reviews.

PetHome — Y&T Paws Pet-Care Management Platform

<https://github.com/xiyunliu610/yt-paws-platform>

Developing a bilingual mobile platform for a real Auckland pet-care business, replacing manual booking, payment and daily care-update workflows.

Tech Stack: React Native (Expo), TypeScript, NestJS, Prisma, PostgreSQL, Stripe

Key Contributions

- Translated real pet-care operations into customer, owner and staff workflows for bookings, pet profiles, service management, daily reports and notifications.
 - Engineered a NestJS and Prisma API with JWT authentication, business-scoped RBAC and overlap-based capacity enforcement, using serializable PostgreSQL transactions to prevent pet, service and staff overbooking.
 - Implemented Stripe Checkout with webhook reconciliation and a manually verified WeChat QR flow, using idempotent operations and PostgreSQL partial unique indexes to prevent duplicate payments across methods.
 - Added English/Chinese localisation, presigned direct-to-object-storage media uploads, in-app notifications and Expo push-delivery infrastructure to the React Native app.
-

EXPERIENCE

Founder & Operations Lead

Oct 2019 - May 2025

Hangzhou Yatian Pet Products Co., Ltd. — Stray Animal Rescue Centre

Key Contributions

- Founded and scaled a stray animal rescue centre supporting the care of up to 300 dogs simultaneously, establishing operational workflows for rescue, rehabilitation and adoption. Mentored volunteers, coordinated veterinary partners and established repeatable operational processes that supported long-term organisational growth, overseeing 1,000+ rescued dogs and 100+ cats.
- Built sustainable operational processes and applied structured decision-making to balance budgeting, logistics, emergency response and animal welfare under significant resource constraints.
- Grew the organisation's social media presence, producing 160+ short-form videos to document rescue work, educate the public and raise awareness of animal welfare.

Software Engineer (Front-end)

Apr 2016 - Oct 2019

Issoftstone Information Technology Co., Ltd.

Key Contributions

- Optimised page load performance by reducing HTTP requests by 40% through CSS sprite sheet optimisation.
- Partnered with backend engineers and product managers to refine requirements, resolve UI issues and deliver new features across multiple enterprise systems.
- Built Vue.js and Element UI interfaces and data visualisations with ECharts, integrating REST APIs across enterprise systems.

VOLUNTEER EXPERIENCE

Software Engineering Volunteer

Jul 2026 - Present

Rural Youth and Adult Literacy Trust (RYALT)

- Reviewed the existing WordPress platform with stakeholders to scope maintainability and scalability improvements.
- Designed a backend refactoring strategy to improve maintainability, scalability and long-term extensibility.
- Contribute to backend redevelopment to improve website performance, user experience and long-term maintainability.

Volunteer (Cashier & Store Assistant)

Aug 2025 - Dec 2025

Salvation Army Family Store

- Operated the POS system, processing cash and card transactions accurately and efficiently. Assisted customers with product selection while maintaining a clean, organised retail environment.

EDUCATION

Master of Information Technology

Expected Completion: 2027

The University of Auckland, New Zealand

- Relevant Coursework: Software Engineering · Web Development · Database Systems · Data Structures & Algorithms

Bachelor of Garden Art Design

2010 - 2014

Zhejiang A&F University (Tianmu College), China

INTERESTS

Building production-oriented software projects, animal welfare, hiking, photography.